Block practice · Logarithms, indices and surds

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Exponential equations, including hidden quadratics

Log equations that turn into a quadratic

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Practice workbook · Exponential equations, including hidden quadratics

Equations with the unknown in a power, solved by matching bases or taking logs, including those that become a quadratic or cubic in a substitution such as u = e^(3x) or u = 2^y.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-LOG-09

1. Question 11 · part a.ii · 3 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part a.ii only.

11

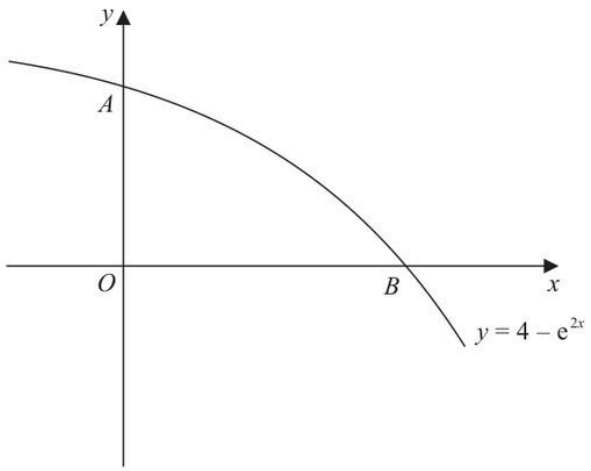


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows part of the curve C with equation $y = 4 - e^{2x}$
The curve C crosses the y-axis at the point A and the x-axis at the point B.

(a) (i) Write down the y coordinate of point A.

(ii) Show that the x coordinate of B is $x = \ln 2$

(3)

The line l is the normal to C at the point B.

(b) Find an equation for l, giving your answer in the form $y = mx + c$

(4)

The finite region R is bounded by C, l and the y-axis.

(c) Using calculus, find the area of R.
Give your answer to one decimal place.

(7)

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1. Question 11 · part a.ii · 3 marks · Working and answer

Working and answer

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2. Question 11 · part a.ii · 3 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a.ii only.

11

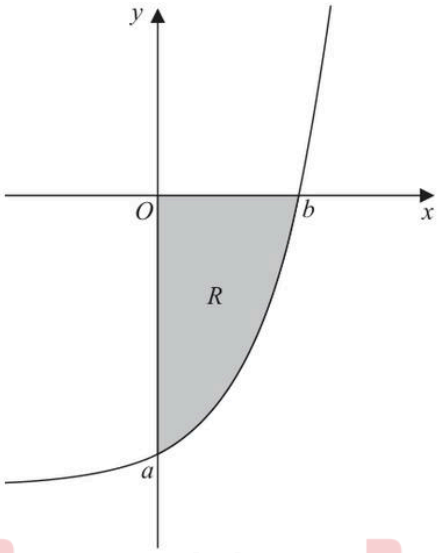


Diagram **NOT** accurately drawn

Figure 3

The finite region R, shown shaded in Figure 3, is bounded by the curve with equation $y = e^{2x} - 9$ and the coordinate axes.

The curve crosses the coordinate axes at the points with coordinates $(0, a)$ and $(b, 0)$

(a) (i) Find the value of a

(ii) Show that $b = \ln 3$

(3)

The region R is rotated through 360° about the x-axis.

(b) Use calculus to find the volume of the solid generated.

Give your answer in the form $\pi(p \ln 3 + q)$, where p and q are integers.

(6)

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32

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2. Question 11 · part a.ii · 3 marks · Working and answer

Working and answer

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3. Question 10 · part a · 4 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

10 (a) Show that $\frac{9^{3y}}{243} = 3^{(6y-5)}$

(4)

(b) Solve the simultaneous equations

$$\frac{9^{3y}}{243} = 27^{(x-2)}$$

$$\log_{10}\sqrt{6xy} = \log_4 2$$

(9)

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3. Question 10 · part a · 4 marks · Working and answer

Working and answer

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4. Question 8 · part a · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

8 The curve C_1 has equation $y = 5e^{-2x} + 4$

The curve C_2 has equation $y = e^{2x}$

The curves C_1 and C_2 intersect at the point A.

(a) Find the exact coordinates of A.

(4)

The tangent at A to C_1 intersects the x-axis at the point B.

(b) Show that the x coordinate of B is $\frac{1}{2}(5 + \ln 5)$

(5)

The tangent at A to C_2 intersects the x-axis at the point D.

(c) Find the area of ΔABD.

(6)

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20

P 6 2 2 8 5 A 0 2 0 3 2

4. Question 8 · part a · 4 marks · Working and answer

Working and answer

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5. Question 8 · part a.i · 0 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

8

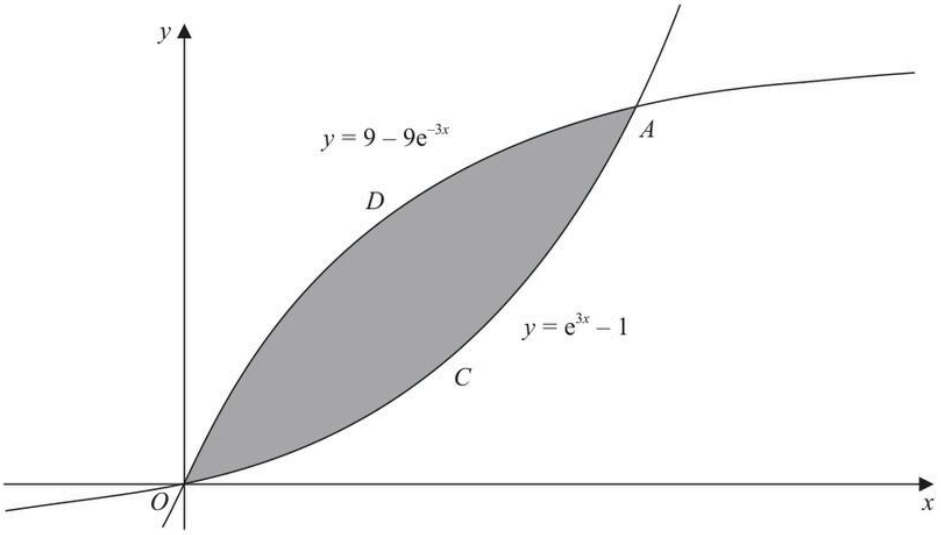


Figure 2

Figure 2 shows part of the curve C with equation $y = e^{3x} - 1$ and part of the curve D with equation $y = 9 - 9e^{-3x}$

The curves intersect at the origin O and the point A.

(a) (i) Show that the x coordinate of the point A satisfies the equation

$$(e^{3x})^2 - 10e^{3x} + 9 = 0$$

(ii) Hence, show that the x coordinate of the point A is $\frac{1}{3}\ln 9$

(5)

The finite region bounded by C and by D is shown shaded in Figure 2.

(b) Use calculus to find the exact area of this region.

(6)

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P 7 1 6 4 2 A 0 2 2 3 6

5. Question 8 · part a.i · 0 marks · Working and answer

Working and answer

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6. Question 8 · part a.ii · 5 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a.ii only.

8

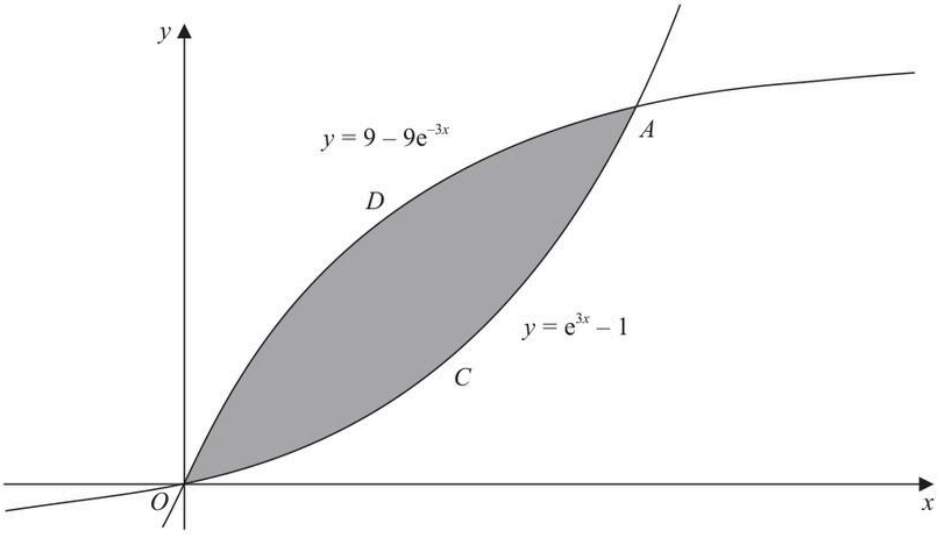


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(6)

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6. Question 8 · part a.ii · 5 marks · Working and answer

Working and answer

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7. Question 5 · part c · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

5 $f(x) = 2ax^3 + x^2 - bx + 3a$ where a and b are integers.

Given that $(x + 2)$ and $(x - 1)$ are both factors of $f(x)$

(a) show that $a = 2$ and find the value of b

(5)

(b) Hence factorise $f(x)$ completely.

(2)

Hence, given that $h(y) = 2^{(3y+2)} + 2^{2y} - 11(2^y) + 6$

(c) solve the equation $h(y) = 0$

Where appropriate give your answers to 3 decimal places.

(5)

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7. Question 5 · part c · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.



7. Question 5 · part c · 5 marks · Working and answer

Working and answer

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Practice workbook · Log equations that turn into a quadratic

Solving equations that mix log_a x and log_x a (or other bases) by changing base so they become a quadratic in one log.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-LOG-11

1. Question 8 · part whole · 7 marks

2020 NOV · Paper 02 (regional R)

Worked-example retry · Attempt part whole only.

8 Solve the equation $\log_3 x - 2\log_x 3 = 1$ (7)

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1. Question 8 · part whole · 7 marks

2020 NOV · Paper 02 (regional R)

Worked-example retry · Attempt part whole only.

Question 8 continued

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(Total for Question 8 is 7 marks)

1. Question 8 · part whole · 7 marks · Working and answer

Working and answer

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2. Question 5 · part b · 7 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

5 (a) Show that $\log_4 32 = \frac{5}{2}$

(2)

(b) Hence, or otherwise, find the exact solutions of the equation

$$\log_2 x - \log_4 32 + \frac{1}{4}\log_x 16 = 0$$

(7)

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P 6 0 4 7 5 A 0 1 0 3 6

2. Question 5 · part b · 7 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 5 continued

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(Total for Question 5 is 9 marks)

2. Question 5 · part b · 7 marks · Working and answer

Working and answer

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3. Question 7 · part c · 6 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

7 A curve C has equation $y = \log_{10}(x + 2)$

(a) Using the axes below, sketch the graph of C.
Label the coordinates of the points of intersection of C with the coordinate axes.

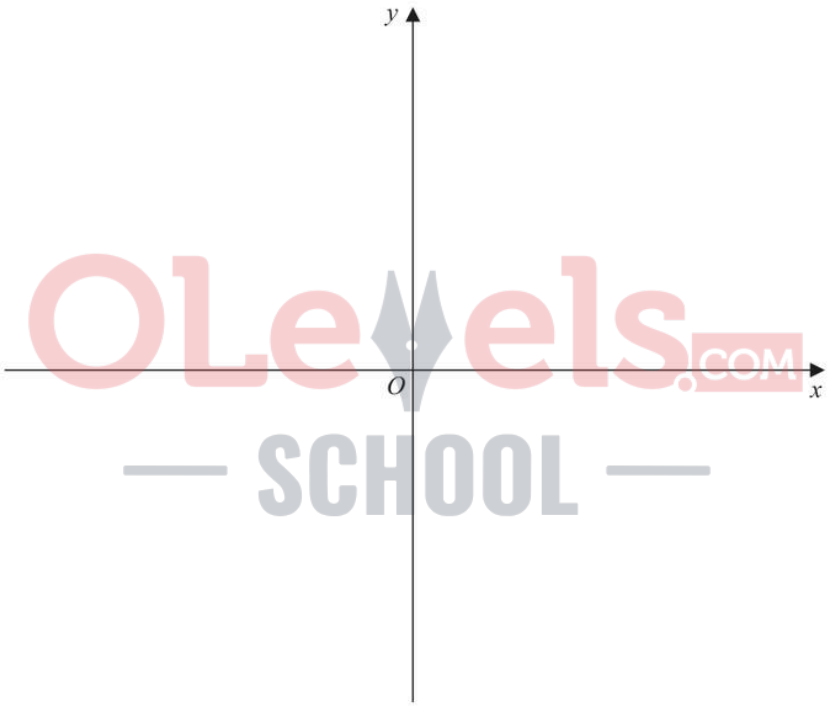
(2)

(b) Solve the equation $2(\log_a 4 + \log_a 16) = 1$

(3)

(c) Solve the equation $5\log_q 16 + 4\log_2 q = 24$

(6)



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3. Question 7 · part c · 6 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 7 continued

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3. Question 7 · part c · 6 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 7 continued

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3. Question 7 · part c · 6 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.



3. Question 7 · part c · 6 marks · Working and answer

Working and answer

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4. Question 6 · part whole · 8 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

6 Solve the equation

$$\log_2 x^3 + \log_4 x^2 - 3\log_x 2 = 0$$

giving your answers to 3 significant figures.

(8)

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4. Question 6 · part whole · 8 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

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Question 6 continued

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(Total for Question 6 is 8 marks)

4. Question 6 · part whole · 8 marks · Working and answer

Working and answer

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5. Question 7 · part b · 5 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

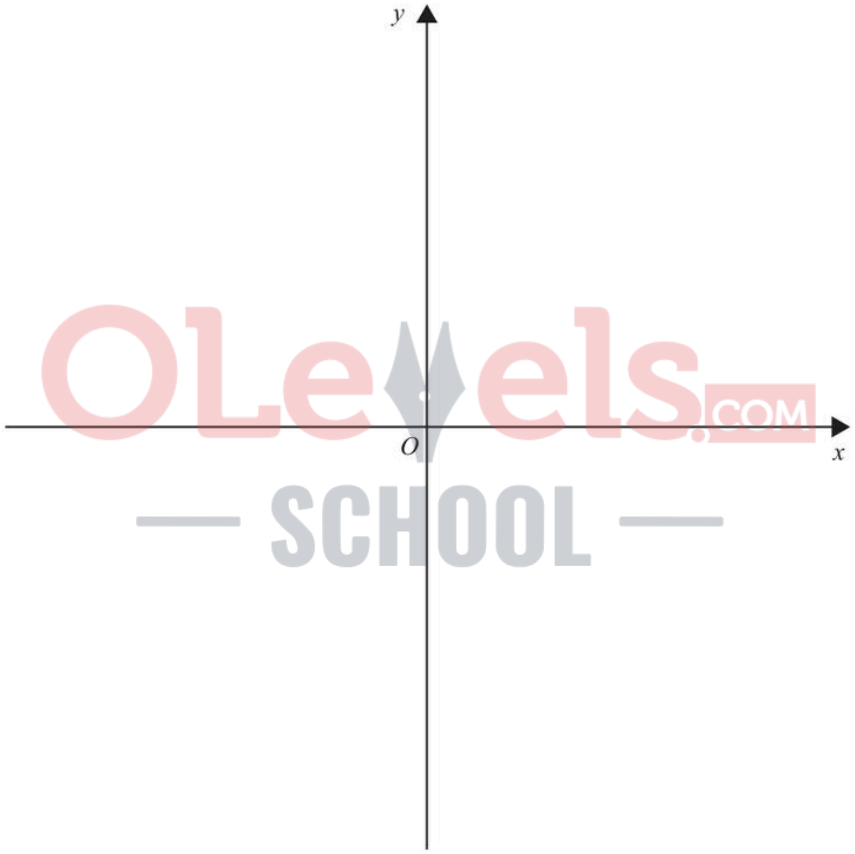
7 The curve C has equation $y = -\log_4(x + 4)$

(a) Using the axes below, sketch the graph of C.
Label the coordinates of the points of intersection of C with the coordinate axes and the equation of any asymptote to C.

(4)

(b) Solve the equation $\log_{(x+4)} 256 - \log_4(x + 4) = 0$

(5)



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5. Question 7 · part b · 5 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 7 continued

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(Total for Question 7 is 9 marks)

5. Question 7 · part b · 5 marks · Working and answer

Working and answer

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6. Question 9 · part ii · 6 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part ii only.

9 (i) Solve the equation $3(\log_a 9 + \log_a 27) = 1$

Give your answer in the form $a = b^c$ where b is prime and c is an integer.

(3)

(ii) Solve the equation $\log_4 p + \log_p 256 = -4$

(6)

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6. Question 9 · part ii · 6 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part ii only.

Question 9 continued

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(Total for Question 9 is 9 marks)

6. Question 9 · part ii · 6 marks · Working and answer

Working and answer

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7. Question 6 · part ii · 7 marks

2024 NOV · Paper 02

Further practice · Attempt part ii only.

6 (i) Solve the equation $5(\log_b 9 + \log_b 3) = 3$

(4)

(ii) Solve the equation $3\log_3 x + 3\log_x 27 = 8\log_4 128$
Give your answers in exact form.

(7)

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7. Question 6 · part ii · 7 marks

2024 NOV · Paper 02

Further practice · Attempt part ii only.

Question 6 continued

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(Total for Question 6 is 11 marks)

7. Question 6 · part ii · 7 marks · Working and answer

Working and answer

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8. Question 10 · part whole · 7 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

10 Solve the equation

$$\log_4 x^3 + 8\log_x 64 = 22$$

(7)

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8. Question 10 · part whole · 7 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 10 continued

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(Total for Question 10 is 7 marks)

8. Question 10 · part whole · 7 marks · Working and answer

Working and answer

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